ZERO IDENTITY · PRIVATE INTENTS

# Private intent.
# Offline relay.
# On-chain proof.

CabalMesh is a zero-identity coordination layer. An intent is signed on-device, carried through nearby peers with no internet, and settled on Avalanche when a gateway returns.

THE GAP

# Coordination assumes a network that is not always there.

Every transaction rail in use today needs two things at once: a live connection and a persistent identity. Both fail in the moments when local coordination matters most.

### Connectivity is brittle

No link, no broadcast. A signed intent has nowhere to go and no path back — it simply fails.

### Identity is exposed

Accounts and public rails leak who acted, from where, and what they intended. The metadata is the leak.

### Settlement needs a witness

A local agreement is worthless to a chain until something proves that it happened, exactly as agreed.

WHAT IT IS

# One device, three layers, no identity.

The stack separates local coordination from global settlement. Each layer is listed with what is running today, not what is intended.

01

## THE CLOAK LAYER

MESH TRANSPORT

libp2p, mDNS discovery and gossipsub over IP; a BLE plane on iOS and Android for the fully offline case. Multi-hop stripping means no hop knows both ends.

`src/mesh.rs` · `crates/cabal-ble` · 9 passing tests

LIVE

02

## THE SETTLEMENT LAYER

AVALANCHE

A minimal Escrow contract, live on Fuji. Offline signing, a local relay queue, and auto-confirmation the moment a gateway appears.

`contracts/Escrow.sol` · `src/blockchain_bridge.rs`

LIVE

03

## THE INVISIBLE BRAIN

LOCAL INTELLIGENCE

A local model parses natural language into a validated intent draft, and Rust still validates and signs it. Zero-knowledge bid proofs and agent-to-agent negotiation are Phase 4 — no proving code is in the build.

`parse_intent_chat` · `commands.rs`

IN PROGRESS

THE CORE LOOP

# Compose. Relay. Reconnect. Settle.

This is the whole product in four moves — and all four are observable in the running app today.

01

### COMPOSE

An intent is written and signed on-device. No account, no server, no session.

02

### RELAY

Nearby peers forward the encrypted payload over BLE or libp2p. No hop learns the origin.

03

### RECONNECT

The first peer holding internet drains the queued work toward the chain.

04

### SETTLE

Escrow confirms on Avalanche Fuji. The result lands on-chain; the identity never does.

A complete, observable lifecycle — not a promise of future infrastructure.

THE DIFFERENTIATOR

# Sign with no network. Confirm when the network returns.

This is the one path nothing else in the category does end to end, and it is the path we demo live.

01

### SIGN OFFLINE

The transaction is signed on-device with the radio down.

`sign_offline`

02

### QUEUE LOCALLY

The signed payload waits in an encrypted local relay queue.

`cabal-vault`

03

### CARRY BY MESH

Peers in radio range move it hop by hop, still offline.

`cabal-ble · mesh.rs`

04

### DRAIN AT GATEWAY

The first peer with internet replays the queue to Avalanche.

`queue replay`

05

### CONFIRM ON-CHAIN

Escrow settles on Fuji and the app auto-confirms the intent.

`deployments/fuji.json`

Verified across two Android emulators with the internet plane disabled — see docs/mobile-build-verification.md.

EVIDENCE

# What is running today.

Audited by reading the code, not the README. Every row names the file it was checked against.

| CAPABILITY | STATUS | CHECKED AGAINST |
|---|---|---|
| Mesh networking — libp2p, mDNS, gossipsub | LIVE | src/mesh.rs · tests/ble_loopback.rs |
| BLE offline plane — iOS and Android | LIVE | src/ble/ · crates/cabal-ble |
| Intent lifecycle — compose, broadcast, settle, proof | LIVE | src/commands.rs · src/intents.rs |
| On-chain Escrow, live on Avalanche Fuji | LIVE | contracts/Escrow.sol · deployments/fuji.json |
| Offline signing, relay queue, auto-confirm | LIVE | src/blockchain_bridge.rs |
| Vault encryption — AES-256-GCM, Argon2id unlock | LIVE | crates/cabal-vault |
| Guardian mesh recovery — enroll, request, approve | LIVE | crates/cabal-guardian · src/guardian.rs |
| Standing — settlement count, computed from chain | LIVE | src/standing.rs |

THE PRIVACY MODEL

# In this network, you are a Nobody.

Privacy in depth is a subtraction, not a feature list. What is removed matters more than what is added.

ERASED

- Physical location — no IP is attached to a relayed intent.
- Account identity — the wallet is created on-device and never registered.
- History — nothing is stored server-side, because there is no server.
- Linkability — guardian requests carry unlinkable per-request tags.

PROVED

- Signature validity — verified by the chain, not by us.
- Escrow state — AVAX locks and releases against the contract.
- Standing — settlement count derived from on-chain history.
- Key custody — hardware-backed on iOS, Android and Apple Silicon.

No zero-knowledge proving ships today — the unused Noir stub was deleted rather than left to imply a capability. A real circuit is Phase 4, labelled, not claimed.

RECOVERY

# Lose the device, keep the wallet.

A wallet with no way out of it is the sharpest risk in any self-custody product. Guardian recovery closes it without ever reconstructing a key on a server.

01

### ENROL

Key shares are split across five chosen guardians. No share is a key.

LIVE

02

### REQUEST

A wiped device broadcasts an unlock request over BLE.

LIVE

03

### APPROVE

Each guardian approves by hand. The reply is gated on a human, never automatic.

LIVE

04

### VETO WINDOW

A 24–48h delay with a push notification, so a stolen device can be blocked.

IN PROGRESS

05

### RESTORE

Three of five shares rebuild the vault on the new device.

LIVE

Passphrase unlock (Argon2id) and full export / import / restore ship today under VAULT → KEYS → ADVANCED.

ARCHITECTURE

# From a dark room to the chain.

Three zones, one direction of travel. Nothing crosses a boundary carrying more than it must.

ZONE 01 · DEVICE

### ON-DEVICE

- Intent composed and signed
- Vault — AES-256-GCM
- Secure element key custody
- Relay queue while offline

ZONE 02 · MESH

### THE MESH

- BLE plane — iOS, Android
- libp2p + mDNS + gossipsub
- Multi-hop metadata stripping
- No hop knows both ends

ZONE 03 · CHAIN

### AVALANCHE FUJI

- Gateway drains the queue
- Escrow locks and releases AVAX
- Settlement proof recorded
- Standing derived on-chain

The mesh never touches the chain. The gateway never learns the origin. The chain never learns the identity.

STATUS BOARD

# What is live, what is not.

A deck that survives questions says this out loud. One true story beats three half-built ones — the offline settlement path is the true one.

### Offline mesh settlement
Sign, relay, confirm — end to end.
LIVE

### BLE plane (iOS · Android)
Two-emulator verification.
LIVE

### Escrow on Fuji
Real deployed address.
LIVE

### Vault + guardian recovery
Enrol, approve, restore.
LIVE

### AI intent parsing
Rust still validates and signs.
IN PROGRESS

### Recovery veto window
Needs device-level push.
IN PROGRESS

### Marketplace + modules
Contracts deployed, UI next.
PLANNED

### ZK bid proofs
Circuit to be written; nargo in CI.
PLANNED

### Agent negotiation · FHE/MPC
Phase 4 and Phase 5.
PLANNED

PLATFORM REALITY

# Where the offline plane actually runs.

The BLE plane needs a radio stack we can drive. Mobile and Apple silicon have it; desktop is scoped, not glossed over.

| PLATFORM | SECURE ELEMENT | BLE PLANE | NOTE |
|---|---|---|---|
| **iOS** | Secure Enclave | YES | Primary target for the offline demo. |
| **Android** | Keystore / StrongBox | YES | Verified across two emulators. |
| **macOS** | Secure Enclave (Apple silicon) | YES | CoreBluetooth; Bluetooth usage declared. |
| **Windows** | TPM | NO | Falls back to the IP plane. |
| **Linux** | Usually none | NO | Falls back to the IP plane. |

The offline plane is scoped where the radio stack is real. Desktop keeps the IP plane — Phase 5 decides the rest.

WHERE IT MATTERS

# Places where the internet is the missing part.

### OFF-GRID AND DISASTER

A cell tower is down or saturated. Value still has to move between people standing near each other, and settle honestly later.

### HYPER-LOCAL TRADE

Markets, borders, festivals, campuses. Counterparties are in radio range; a payment rail that needs a data plan is not.

### PRIVACY-CRITICAL WORK

Journalists, field researchers, activists. The metadata trail is the risk, so the network is built to leave none.

The same primitive serves all three: a signed intent that survives without a network and proves itself when one returns.

HARDWARE · PLANNED

# Two boxes the software would run on.

Neither exists — no silicon selected, no board, no tooling. Both are drawn to the millimetre from workloads the app already performs, so the spec can be checked against the code instead of a mood board.


PLANNED

## SHADOWBOX

MESH · MODEL · PROVER

220 × 160 × 70 mm · 1.9 kg · fanless

One node instead of three: it relays for the neighbourhood, runs the local model that reads a sentence, and generates the proof. RADIO / CRYPTO / POWER bays mirror the in-app module system.

PLANNED

## THE NOBODY BOX

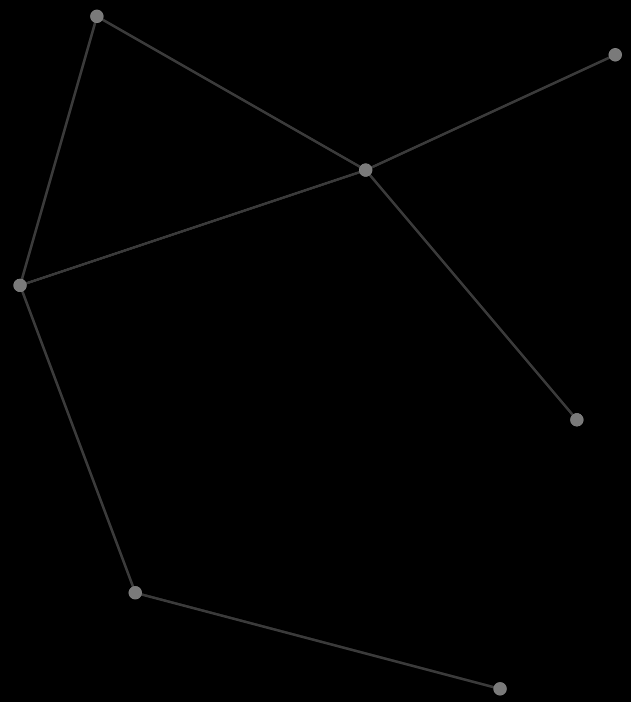
PHYSICAL ESCROW

520 × 460 × 720 mm · ≈ 88 L · no keyhole

A parcel locker whose bolt turns on the on-chain release. The seller drops the item in, the buyer opens it, nobody in between can. A load cell reports that something left — never what it was.

SPEC AND DRAWINGS — docs/pitch/hardware-device-prompts.md · the images above are vector renders of that geometry, not photographs

# Zero identity.
# Private intents.
# Verifiable execution.



| | |
|---|---|
| DEMO | youtu.be/Z3ooub-mnCw |
| REPOSITORY | github.com/kurodenjiro/Cabal-Mesh |
| NETWORK | Avalanche Fuji — Escrow live |
| STATUS | Offline settlement path: shipped and demonstrable |

WE LEAVE NO IDENTITY, ONLY TRACES.